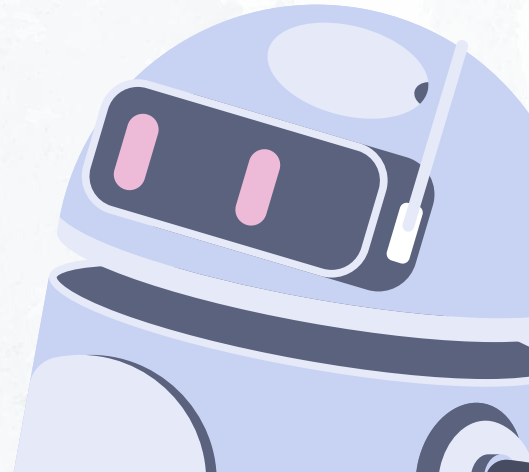


Super-Resolution of Medical MNIST Images Using WGAN

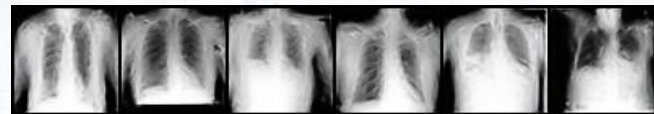
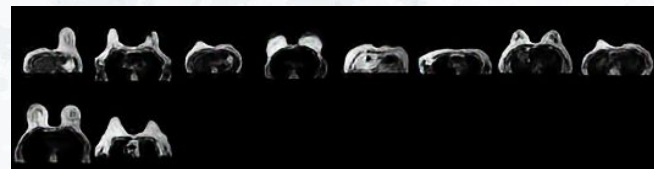
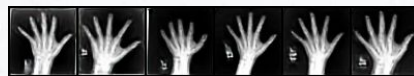
Gianmaria Pizzo (872966)



The Super Resolution Task (SR)

A computer vision task whose **objective** is to produce an output with a **higher resolution** than the input, while **maintaining the original content and structure**.

In our case, we challenge a model with low resolution (downscaled) **28x28 medical images** [5] and hope to get a 64x64 version of them.



28x28 downsampled images

64x64 images generated by our WGAN

But first...

Ingredients

For our goal we are going to cover 3 core aspects that make our project effective:

- **Generative Upscaling Architecture:** We need to build a network that is capable of **creating sharp details** through feature extraction and upscaling. This justifies our use of a GAN with **convolutional and upscaling layers** [1].
- **Effective Loss Functions:** The training procedure needs to be as stable as possible for our objective. Thus, we combine the **Wasserstein distance** [3] and **Gradient Penalty** [2] for the critic's loss.
- **Yield of Realism:** The generator needs to learn **realistic aspects** of the input images. Thus, we combine the **Adversarial and Pixel-wise L1 losses** for a better direction. Furthermore we do not feed noise, like regular GANs, but we use **downscaled versions of the original images** and we tune parameters to avoid overfitting.

Generative Adversarial Networks (WGANs)

A GAN is a generative model that **simultaneously trains two models**:

- The Generator or **G**, which generates a data distribution and “*tries to fool D*”.
- The Discriminator or **D**, that “estimates the probability that a sample came from the training data rather than **G**” [4].

Both networks compete with each other through minimax optimization of a distance **D**, like the KL divergence or the Jensen-Shannon divergence

$$\min_G \max_D \mathbb{E}_{x \sim \mathbb{P}_r} [\log(D(x))] + \mathbb{E}_{\tilde{x} \sim \mathbb{P}_g} [\log(1 - D(\tilde{x}))],$$

The a goal to reach an equilibrium state, called **Nash Equilibrium**, which would guarantee (in theory) the strengthening of both networks to bring forth realistic HR image.

However ...

Issues in GANs

Although GANs has been shown to achieve great realistic image generation, their **training is not trivial**; The process is known to be slow and unstable, due to the following aspects

- **Nash Equilibrium**: Both models learn independently, so achieving it is hard and requires careful tuning.
- **Vanishing Gradient**: If **Discriminator** evolves to be **too strong** (converges too quickly), the gradient of the **generator's loss vanishes** and prevents learning [2].
- **Mode Collapse**: GANs sometimes **fail to generate diverse samples** and collapse into producing only a few modes of the data distribution (e.g., a GAN trained on faces might generate only a few similar-looking faces) [2].



Use **Wasserstein GANs** and **Gradient Penalty**!

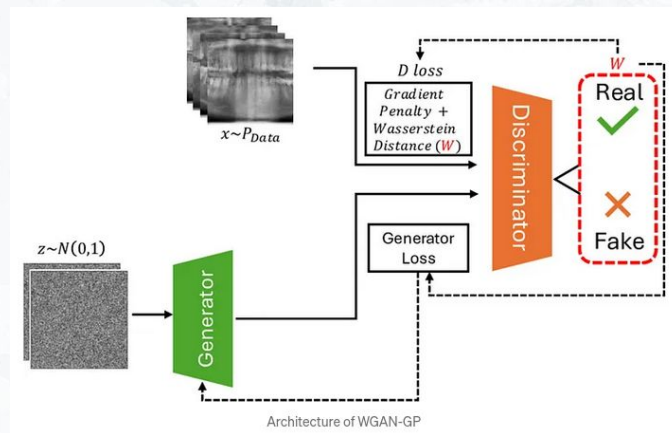


Image from [3]. The only difference is that our input is made of low res images (for the generator) and the critic receives both fake hr and real hr images.

Wasserstein Distance as Critic loss function

For discrete probability distributions, the Wasserstein distance is also called the **earth mover's distance (EMD)** [3]. If we imagine the distributions as different heaps of a certain amount of earth, then the EMD is the **minimal total amount of work it takes to transform one heap into the other**. Since computing the primal EM distance is **intractable**, we use the derivation obtained through the KR duality.

$$W(\mathbb{P}_r, \mathbb{P}_g) = \inf_{\gamma \in \Pi(\mathbb{P}_r, \mathbb{P}_g)} \mathbb{E}_{(x,y) \sim \gamma} [\|x - y\|] , \quad \xrightarrow{\text{Kantorovich-Rubinstein duality [2]}} \quad W(\mathbb{P}_r, \mathbb{P}_\theta) = \sup_{\|f\|_L \leq 1} \mathbb{E}_{x \sim \mathbb{P}_r} [f(x)] - \mathbb{E}_{x \sim \mathbb{P}_\theta} [f(x)]$$

Let us observe that the critic's objective is to approximate the real Wasserstein distance and learn how to **separate real images from fake ones**. Then, we can work on a **minimization problem**:

$$\max_{f_w, \|f\|_L \leq 1} \left[\mathbb{E}_{x \sim P_r} [f_w(x)] - \mathbb{E}_{x \sim P_g} [f_w(x)] \right] \quad \longrightarrow \quad \min_w - (\text{mean}(f_w(\text{real})) - \text{mean}(f_w(\text{fake})))$$

But we **still need to enforce 1-Lipschitz constraint** that requires the function learnt by the **critic** to be smooth (avoid abrupt changes and steep curves that lead to unstable gradients).

WGAN-GP: Gradient Penalty

This can be achieved by **Gradient penalty**. How? We ensure the critic **behaves linearly** (or nearly so) between real and fake samples, since that is **where most of the generator's training signal comes from**.

First, we sample some images by randomly interpolating between real and fake examples.

$$\hat{x} = \epsilon x_{\text{real}} + (1 - \epsilon)x_{\text{fake}}, \quad \epsilon \sim U[0, 1]$$

The critic assigns higher scores to real images and lower scores to fakes, forming a **scalar “slope”** between them. The **steepness** of this slope is **captured by the gradient of the critic's output with respect to its input** and determines the quality of the learning signal passed to the generator. Thus, we compute

$$\text{grad} = \nabla_{\hat{x}} f(\hat{x}) \Rightarrow \|\text{grad}\|_2 = \|\nabla_{\hat{x}} f(\hat{x})\|_2$$

Finally, to guarantee a smooth and stable gradient (i.e., no exploding or vanishing values), we enforce that the **critic's gradient norm remains close to 1** along the lines connecting real and fake samples.

This results into the following combined loss function:

$$\mathcal{L}_{\text{critic}} = - \left(\mathbb{E}_{x \sim P_r} [f_w(x)] - \mathbb{E}_{x \sim P_g} [f_w(x)] \right) \cdot \lambda_{was} + \mathbb{E}_{\hat{x}} \left[(\|\nabla_{\hat{x}} f_w(\hat{x})\|_2 - 1)^2 \right] \cdot \lambda_{gp}$$

Focus on the generator loss

Since the generator is **not just trying to “fool”** the critic, it’s also expected to produce output structurally close to the real target, we cannot just rely on adversarial loss.

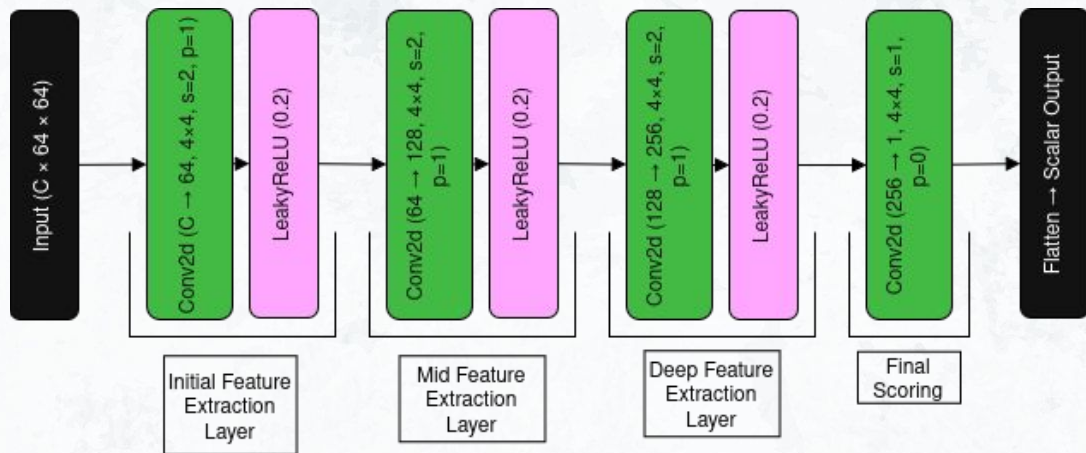
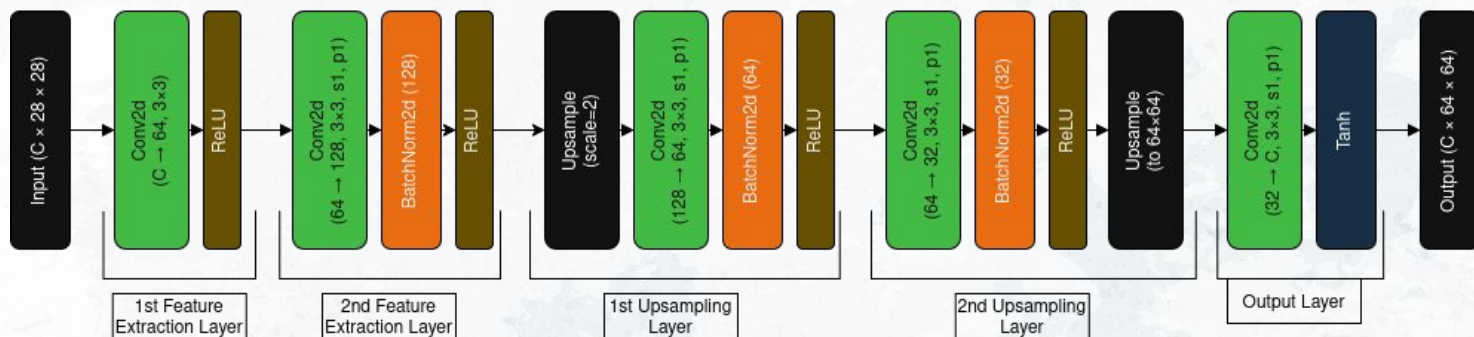
Adversarial loss surely encourages **realism**, as the **generator wants the critic to assign high scores to fakes**. Therefore, the output will be sharp but we have **no certainty that it will grasp the features** of the images because it relies on what the “critic wants to see”.

On the other hand, **Pixel-wise L1** loss aims to deliver **accuracy**. It forces generated images to be **close to the ground truth**, preserving layout and structure. However, it **penalizes every pixel independently** and by itself can lead to blurry images.

We combine these measures, along with some constants, to get a good tradeoff.

$$\mathcal{L}_{\text{gen}} = \underbrace{-\mathbb{E}[f_w(g_\theta(x_{\text{low}}))]}_{\text{Adversarial(WGAN)}} \cdot \lambda_{\text{adv}} + \underbrace{\|g_\theta(x_{\text{low}}) - x_{\text{high}}\|_1}_{\text{Pixel-wise L1 loss}} \cdot \lambda_{\text{L1}}$$

Super Resolution WGAN-GP Architecture



Methods and Experiments:

- We use a **configuration module** that ensures to load all the parameters needed for the training/testing procedure and to target the right dataset. This was initially extended to combine different pairs of generators and critics, along with different losses and many other aspects.
- The regular MNIST dataset was not a real challenge for the model, so **we switched to more complex images** with 3 channels and hard-to-catch details. We focused our efforts on 3 different subsets of the Medical MNIST dataset [5]: “BreastMRI”, “CXR”, “Hand”. Each subset is made of 10.000 images which are divided into 9.000 train images and 1.000 test images.
- The learning process is dealt with **small batches** of 16 images and very **small learning** rates to avoid overfitting. We track everything through **Tensorboard** and visualize the training and validating steps live from the browser.
- The **test can be done separately with a dedicated module** that automatically picks up the state of the models, the test data and runs the test.
- All the evaluations are based both on the user perception and metrics like **MSE, RMSE, PSNR, SSIM, LPIPS and FID**.

Demo and Results

Let us dive into the code for a nice demo and show some pre-trained models

Conclusions

1. The training becomes immediately stable when we followed the directions suggested by the papers concerning WGAN [3] and WGAN-GP [2].
2. Although having a simple architecture, and being trained on limited examples, the generator behaves well. In particular, its performances skyrocket when the images contain distinguished colors.
3. The generator tends to forget about the details (for example ribs in Chest XRays). Like the SR-GAN paper suggests [1], we should be either using some deeper architecture or modify the loss to include a perspective that really pushes the generator to capture more details (for example VGG + perceptual loss).
4. The critic often dominates over the generator, which indicates the need for a more careful tuning of the parameters
5. The models can bend very well to a simple super resolution job, and can be improved to deliver higher resolution in a modular way.

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